

Daniil Makeev

Senior AI Engineer · LLM Engineer

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Yerevan, Armenia · Open to relocation / remote · EAEU work authorization

Senior AI/LLM Engineer with 5 years of software engineering background, specializing in **production LLM systems**, agentic workflows, RAG, evaluation, and resilient AI infrastructure. Designed and shipped multi-agent systems, multimodal generation pipelines, retrieval infrastructure, and stateful AI services with measurable results: **6x faster** processing, **–60% generation cost**, and database calls cut from **3,701 to 22**. Hands-on with LangGraph, LangChain, DeepAgents, MCP, hybrid retrieval, pgvector, structured outputs, tool use, context engineering, model routing, async Python, LLMops, and production integration of non-deterministic AI providers.

EXPERIENCE

hao.vc (AI Lab) — Senior Applied AI Engineer / Tech Lead

Jan 2025 – Present

Project: Autonomous Multimodal Generation Platform & LLMops Engine

- Designed and productionized an end-to-end multimodal AI generation platform in **Python** and **Prefect**, orchestrating LLM-based script generation, voice synthesis, image generation, and media composition across **OpenAI API, ElevenLabs, Google Imagen, FFmpeg**, and **AWS S3**; reduced end-to-end latency **6x (~60 → ~10 min)** and generation cost **–60% (\$2.00 → \$0.80)**.
- Built the Resilience & LLMops layer around non-deterministic AI providers: exponential-backoff retries, fallback routing, automated media validation, and auto-repair of malformed structured outputs with **Pydantic v2**; covered production workflows with **200+ automated tests**.
- Supported controlled **A/B experimentation** of generation features through runtime configuration and feature switches, so media-generation behavior could be tested without rebuilding the pipeline.
- Owned architecture, provider budgets, and production delivery, shipping **100+ AI-generated media assets** that accumulated **800K+ organic views**.

Project: Browser AI Agent Framework

- Architected a **multi-agent framework** using an orchestrator-worker model with subagents and router / planner-executor patterns on **MCP**, orchestrated through **LangGraph.js / LangChain.js / ReAct**; agents performed autonomous tool use, DOM interaction, document processing, and task execution via function calling and structured outputs.
- Built the Evals & Safety layer: **LLM-as-a-judge** regression evaluation, recurring failures classified into **6 error classes**, degradation controls, prompt-injection defenses, execution isolation, graph-level observability, token telemetry, and **200+ automated tests** across agent iterations.
- Designed the browser-agent runtime on **Chrome MV3** with **Human-in-the-Loop** controls for sensitive and non-standard actions.

Project: RAG, Memory & Model-Routing Infrastructure

- Built an end-to-end **RAG** and semantic-memory pipeline (ingestion, chunking, embeddings, retrieval, generation) on **PostgreSQL / pgvector**, with hybrid dense + sparse/full-text retrieval, **RRF**, and **cross-encoder reranking** for semantic search over the knowledge base.
- Optimized retrieval and query execution, reducing database calls on critical paths from **3,701 to 22 (–99.4%)**.
- Implemented conditional **model routing** across Claude, DeepSeek, and Qwen based on task complexity and pipeline stage, optimizing inference cost while reserving higher-capability models for agentic workloads.

Project: AI News Intelligence Platform

- Built a stateful agentic workflow on **DeepAgents / LangGraph** with tenant-scoped tool use, conversational memory, and context compaction; explicitly separated deterministic LLM pipelines from model-driven agent loops to keep execution bounded and observable.
- Designed **prompt- and context-engineering** and degradation controls for long-running sessions: summarized historical context, preserved tool-call / tool-result boundaries, and compacted conversation state without corrupting agent execution.
- Implemented tool-driven agent execution for searching historical content, retrieving recent context, and reading user-specific state through isolated tenant-scoped capabilities.
- Built **token- and inference-cost observability** across LLM stages, aggregating prompt/completion tokens, model calls, and provider-reported USD cost to compare model configurations and control LLM spend.

- Operated the system as a **multi-tenant AI service** with per-user model selection, personalization, quotas, isolated history, and shared production pipelines.

Project: EdTech Interview-Scheduling Platform

- Built and delivered a production backend for an interview-scheduling platform using **FastAPI**, **PostgreSQL/asyncpg**, and HMAC-based authentication, serving **~700 active students**.
- Implemented end-to-end delivery infrastructure: lint/test gates, **Docker/GHCR** builds, automated deployment, and versioned SQL migrations on client infrastructure.

Zenit-Electro — Backend / Applied AI Engineer

Jun 2023 – Dec 2024

- Designed and maintained **Python/SQL** data-automation pipelines for catalog import/export, validation, normalization, and internal operational workflows, saving non-engineering teams an estimated **~8–12 manual hours per week**.
- Expanded from deterministic backend automation into applied AI workflows, introducing **LLM-assisted processing**, structured model outputs, and retrieval-oriented prototypes for semi-structured business data.
- Worked across the full automation lifecycle — ingestion, validation, transformation, persistence, and internal tooling — building the backend/data foundation for later production LLM and agentic systems.

EPAM Systems — Software Engineer (Backend)

Oct 2021 – Jun 2023

Middle Backend Engineer

Jun 2022 – Jun 2023

- Designed and maintained scalable **REST microservices** with **FastAPI**, **PostgreSQL**, and **asyncio** for enterprise workloads; optimized SQL queries and indexes for high concurrency and backend throughput.
- Built asynchronous data-processing and integration pipelines, handling service-to-service communication, validation, retries, and data transformation across distributed backend components.
- Contributed to classical **NLP** workflows using Python, spaCy, and transformers/BERT for enterprise text extraction and categorization.
- Maintained CI/CD pipelines with **Docker**, **GitHub Actions**, and **pytest**; participated in code reviews and engineering standards across multidisciplinary teams.

Intern / Junior Python Engineer

Oct 2021 – May 2022

- Developed and maintained Python backend services, REST APIs, PostgreSQL integrations, and asynchronous application logic.
- Built basic ETL/data-processing workflows, service integrations, automated tests, and CI/CD pipelines.
- Progressed from implementation-level backend tasks into ownership of production services and distributed backend components.

CORE SKILLS

AI / LLM: Production LLM Systems, Agentic Workflows, Multi-Agent Systems, Orchestrator-Worker, Subagents, Router / Planner-Executor, LangChain, LangGraph, LangGraph.js, DeepAgents, MCP, ReAct, Tool Use, Function Calling, Structured Outputs, Prompt Engineering, Context Engineering, Memory / Compaction

RAG / Retrieval: End-to-End RAG Pipelines, Ingestion, Chunking, Embeddings, pgvector, Dense + Sparse / Full-Text Hybrid Retrieval, RRF, Cross-Encoder Reranking, Semantic Search

Evals / LLMops: LLM-as-a-judge, Regression Evaluation, Degradation Control, Failure Recovery, Prompt-Injection Defenses, Token / Cost Observability, Model Routing, Inference Cost Optimization, LLM Reliability

Models / AI Tooling: OpenAI API, Anthropic Claude API, Google Gemini, OpenRouter, DeepSeek, Groq Whisper, ElevenLabs, Prefect, FFmpeg/ffprobe

Backend / Platform: Python, TypeScript, SQL, FastAPI, REST, Pydantic v2, PostgreSQL, asyncpg, asyncio, Supabase, AWS S3, Docker, GitHub Actions, GHCR, React, Chrome Extension MV3, Telegram Mini Apps, Linux

EDUCATION & LANGUAGES

Bauman Moscow State Technical University (BMSTU) — BSc Computer Engineering / Computer Systems ·

Languages: Russian — native · English — B2 (Upper-Intermediate)